PHYS 209: Aspects of Modern Physics

Module 4: Qubits

Lecture 7: Quantum Error Correction

Instructor: Muhammad Hani Zaheer

Review

- Quantum advantage requires large number of qubits.
- Due to decoherence from environmental interactions, qubits lose phase information.
- $\widehat{H}_{int} = g\hat{Z} \otimes \hat{Z}^{\otimes N}$:
  - $N = 1$: Oscillating coherences between the system and environment qubits.
  - $N \to \infty$: Decaying coherences.
- $\widehat{H}_{int} = g\hat{X} \otimes \hat{X}$:
  - $N = 1$: System qubit oscillates between the two z eigenstates [Bit flip].
- These environmental effects manifest as errors in computations.
- Need a way to deal with errors.

Qubit Errors

- An error manipulates qubit state $|\psi\rangle$.
  - Hence, it acts as an operator acting on the qubit.
- A general quantum operation acts on density matrix $\hat{\rho}$ and can be written as

$$\hat{\rho} \rightarrow \sum_a \hat{K}_a \hat{\rho} \hat{K}_a$$

  - Kraus operators $\hat{K}_a$ satisfy $\sum_a \hat{K}_a^\dagger \hat{K}_a = 1$.
  - $\{\hat{I}, \hat{X}, \hat{Y}, \hat{Z}\}$ forms a basis for operators in 2 dimensions.
  - Hence, $\hat{K}_a$ can be expanded in terms of Pauli matrices:

$$\hat{K}_a = a\hat{I} + b\hat{X} + c\hat{Y} + d\hat{Z}$$

- Therefore, any quantum channel is a probabilistic mixture of $\hat{X}, \hat{Y}, \hat{Z}$.
- This justifies treating continuous noise and discrete Pauli errors accumulated in short time intervals.
  - Discretization of errors is key to quantum error correction.

Bit Flip Error

- Consider a qubit-qubit interaction

$$\hat{H}_{int} = \hbar g \hat{X} \otimes \hat{X}$$

  - For qubit starting in $|0\rangle$ state, this interaction leads to a classical mixture

$$\hat{\rho}_1(t) = \begin{bmatrix} \cos^2(gt) & 0 \\ 0 & \sin^2(gt) \end{bmatrix} = \cos^2(gt)\,|0\rangle\langle 0| + \sin^2(gt)\,|1\rangle\langle 1|$$

  - At times $t = \frac{(2n+1)\pi}{2g}$ for $n = 0,1, \ldots,$ → the qubit state flips to $|1\rangle$.

- The Pauli matrix $\hat{X} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ models a bit flip error:

$$\hat{X}(\alpha|0\rangle + \beta|1\rangle) = \alpha|1\rangle + \beta|0\rangle$$

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \begin{bmatrix} \beta \\ \alpha \end{bmatrix}$$

  - The logical values 0 and 1 get swapped.

- For a probability p of bit flip, the density matrix evolves as

$$\hat{\rho} \to (1-p)\hat{\rho} + p\hat{X}\hat{\rho}\hat{X}$$

Classical Error Correction

- Classically, we can protect bits by using multiple copies

$$0 \to 000, \qquad 1 \to 111$$

- If one bit flips, majority still preserves original bit value

$$001 \to 0, \qquad 010 \to 0, \qquad 100 \to 0$$
$$110 \to 1, \qquad 101 \to 1, \qquad 011 \to 1$$

- Can we do the same for quantum bits?
  - No: No Cloning Theorem.
    - Can copy along a basis. For example, can copy $|0\rangle, |1\rangle$.
    - Cannot copy general qubit state. For example, cannot copy $\alpha|0\rangle + \beta|1\rangle$ for general α, β.
- Idea:
  - Use entanglement to encode bit information in two ancilla qubits.

Encoding the Logical Qubit

- Logical qubits are encoded as

$$|0\rangle_L = |000\rangle, \qquad |1\rangle_L = |111\rangle$$

- For a general qubit state,

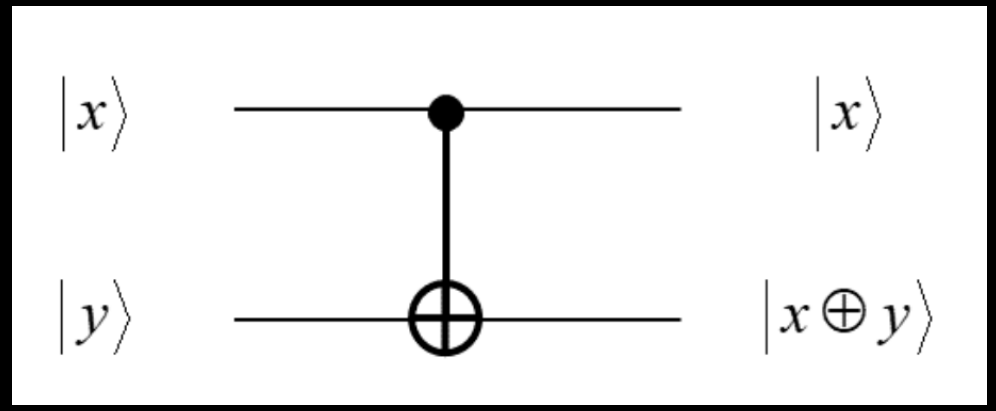
$$\alpha|0\rangle + \beta|1\rangle \rightarrow \alpha|000\rangle + \beta|111\rangle$$

- Question: How to use the two ancillas to do this?
  - Start with both ancillas at $|0\rangle$.
  - If logical qubit is $|0\rangle$, do nothing to ancillas.
  - If logical qubit is $|1\rangle$, flip ancillas to $|1\rangle$.

} CNOT Gate

CNOT Gate

Control $\|x\rangle$	Target $\|y\rangle$	Output $\|x \oplus y\rangle$
$\|0\rangle$	$\|0\rangle$	$\|0\rangle$
$\|0\rangle$	$\|1\rangle$	$\|1\rangle$
$\|1\rangle$	$\|0\rangle$	$\|1\rangle$
$\|1\rangle$	$\|1\rangle$	$\|0\rangle$



- Matrix representation:

$$CNOT = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

- It is used to produce Bell states.
- CNOT + single-qubit gates = universal set of quantum gates.

How CNOT Gate encodes logical bit

- Start with logical qubit state $\alpha|0\rangle + \beta|1\rangle$ and ancilla state $|00\rangle$

$$(\alpha|0\rangle + \beta|1\rangle) \otimes |00\rangle = \alpha|000\rangle + \beta|100\rangle$$

- Apply CNOT with qubit 1 as control and qubit 2 as target

$$\alpha|000\rangle + \beta|100\rangle \rightarrow \alpha|000\rangle + \beta|110\rangle$$

- Apply CNOT with qubit 1 as control and qubit 3 as target

$$\alpha|000\rangle + \beta|110\rangle \rightarrow \alpha|000\rangle + \beta|111\rangle$$

- We have successfully encoded the **logical qubit** into three **physical qubits**.

The Error Model

$$\alpha|0\rangle + \beta|1\rangle \rightarrow \alpha|000\rangle + \beta|111\rangle$$

- Assume that at most one of the three qubits flips.
  - Valid when error probability p is small.
- Four possibilities:
  - No error: $\alpha|000\rangle + \beta|111\rangle$
  - Flip qubit 1: $\alpha|100\rangle + \beta|011\rangle$
  - Flip qubit 2: $\alpha|010\rangle + \beta|101\rangle$
  - Flip qubit 3: $\alpha|001\rangle + \beta|110\rangle$
- These four states are mutually orthogonal.
  - So, they are perfectly distinguishable without disturbing α or β.

Detecting Error without reading the State

$$\alpha|000\rangle + \beta|111\rangle, \qquad \alpha|100\rangle + \beta|011\rangle$$
$$\alpha|010\rangle + \beta|101\rangle, \qquad \alpha|001\rangle + \beta|110\rangle$$

- If we measure the qubit, the superposition collapses.
  - This will lose α, β.
- Measure parities instead:
  - $\hat{M}_1 = \hat{Z} \otimes \hat{Z} \otimes \hat{I}$ Do qubits 1 and 2 agree?
  - $\hat{M}_2 = \hat{I} \otimes \hat{Z} \otimes \hat{Z}$ Do qubits 2 and 3 agree?
- These are called syndrome measurements.
  - Measuring $+1$ Qubits agree No error
  - Measuring -1 Qubits disagree Error

Reading the Syndrome

State	$\hat{M}_1 = \hat{Z} \otimes \hat{Z} \otimes \hat{I}$	$\hat{M}_2 = \hat{I} \otimes \hat{Z} \otimes \hat{Z}$	Error
$\alpha\|000\rangle + \beta\|111\rangle$	$+1$	$+1$	None
$\alpha\|100\rangle + \beta\|011\rangle$	-1	$+1$	Qubit 1
$\alpha\|010\rangle + \beta\|101\rangle$	-1	-1	Qubit 2
$\alpha\|001\rangle + \beta\|110\rangle$	$+1$	-1	Qubit 3

- Each error maps to a unique syndrome.
- Measuring the parity pair $\hat{M}_1, \hat{M}_2$ tells us which qubit flipped.

Correcting the Error

State	$\hat{M}_1 = \hat{Z} \otimes \hat{Z} \otimes \hat{I}$	$\hat{M}_2 = \hat{I} \otimes \hat{Z} \otimes \hat{Z}$	Error	Apply
$\alpha\|000\rangle + \beta\|111\rangle$	$+1$	$+1$	None	$\hat{I} \otimes \hat{I} \otimes \hat{I}$
$\alpha\|100\rangle + \beta\|011\rangle$	-1	$+1$	Qubit 1	$\hat{X} \otimes \hat{I} \otimes \hat{I}$
$\alpha\|010\rangle + \beta\|101\rangle$	-1	-1	Qubit 2	$\hat{I} \otimes \hat{X} \otimes \hat{I}$
$\alpha\|001\rangle + \beta\|110\rangle$	$+1$	-1	Qubit 3	$\hat{I} \otimes \hat{I} \otimes \hat{X}$

- In either case, we recover:

$$\alpha|000\rangle + \beta|111\rangle$$

- We did not measure the state or affect α, β.
- We used the syndrome to identify error and corrected it.

Correcting Phase Flip Errors

- Three-bit bit-flip code allows us to correct bit flips $\hat{X}$.
- Phase flip $\hat{Z}$ acts of $|+\rangle, |-\rangle$ analogously to how $\hat{X}$ acts on $|0\rangle, |1\rangle$.

$$\hat{Z}|+\rangle = |-\rangle, \qquad \hat{Z}|-\rangle = |+\rangle$$
$$\hat{X}|0\rangle = |1\rangle, \qquad \hat{X}|1\rangle = |0\rangle$$

- Another perspective: $\hat{X}$ and $\hat{Z}$ are related by a basis change $\hat{H}\hat{Z}\hat{H} = \hat{X}$.
- To deal with phase flips ($\hat{Z}$ errors):
  - Change basis from $|0\rangle, |1\rangle$ to $|+\rangle, |-\rangle$.
  - Apply three-bit bit flip code.
  - This corrects phase flip errors.

Correcting $\hat{Y}$ Errors

- $\hat{Y} = i\hat{X}\hat{Z}$
- Check:

$$i\hat{X}\hat{Z} = i\begin{bmatrix}0 & 1\\ 1 & 0\end{bmatrix}\begin{bmatrix}1 & 0\\ 0 & -1\end{bmatrix} = i\begin{bmatrix}0 & -1\\ 1 & 0\end{bmatrix} = \begin{bmatrix}0 & -i\\ i & 0\end{bmatrix} = \hat{Y}$$

- So, $\hat{Y}$ error is combination of bit flip $\hat{X}$ and phase flip $\hat{Z}$.
  - Hence correcting $\hat{X}, \hat{Z}$ errors also corrects $\hat{Y}$.

Correcting All Errors

- A universal QEC code only needs to correct $\hat{X}, \hat{Z}$.
  - Naïve approach:
    - $3 \times 3 = 9$ physical qubits to protect 1 logical qubit → Shor code.
    - Produced by concatenating bit flip and phase flip codes.
  - Most efficient code:
    - 5 perfect qubits are enough → Theoretical minimum set by Quantum Hamming Bound.
- Practically:
  - Current technology has error rates that are too high.
  - Cannot assume single error at a time.
  - Need more physical qubits – 100-1000 per logical qubit.

Other Quantum Error Correction Schemes

- Surface Codes: Encode logical qubit onto 2-dimensional lattice of physical qubits and check errors in neighboring qubits.
  - Pursued by Google and IBM.
- Quantum LDPC (Low Density Parity Check): Requires long-range interactions but fewer physical qubits per logical qubit.
  - Pursued by IBM.
- Bosonic Codes: Encode logical qubit onto harmonic oscillator.
  - Pursued by AWS, Alice & Bob, and Nord Quantique.
- Topological Qubits: Hardware-level protection of errors by encoding information non-locally.
  - Pursued by Microsoft.

Outlook

- Current: NISQ (Noisy Intermediate-Scale Quantum) era
  - Qubits are noisy → error rates are high.
  - Long algorithms are unfeasible due to noise accumulation.
- The Threshold Theorem:
  - If physical error rate per gate < critical threshold, errors can be corrected faster than they accumulate.
  - This enables arbitrarily long computation.
- Future: FTQC (Fault Tolerant Quantum Computation)
  - Error suppression + error correction.
  - Allow arbitrarily long computation running long algorithms.

Qubits in Industry

Platform	Key Players
Superconducting	Google, IBM, Rigetti
Trapped Ion	IonQ, Quantinuum
Neutral Atom	Atom Computing, PASQAL
Photonic	Xanadu
Topological	Microsoft
Bosonic	AWS, Alice & Bob

- No single platform has won.
- QEC is common bottleneck.

Review of Module

- Single qubit: Use superposition
  - As computational resource.
  - For sensing (Ramsey sequence).
- Two qubits use Bell states (entanglement) for
  - Tests of local realism
  - Quantum cryptography
- Many qubits can be used for
  - Quantum algorithms
  - Modeling quantum noise (decoherence)
  - Protecting qubits from noise (quantum error correction)